Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Generating a gene marker panel to identify murine macrophage polarisation subtypes

Finnegan Á[1], Smith A[2], Creagh E[3]

Introduction

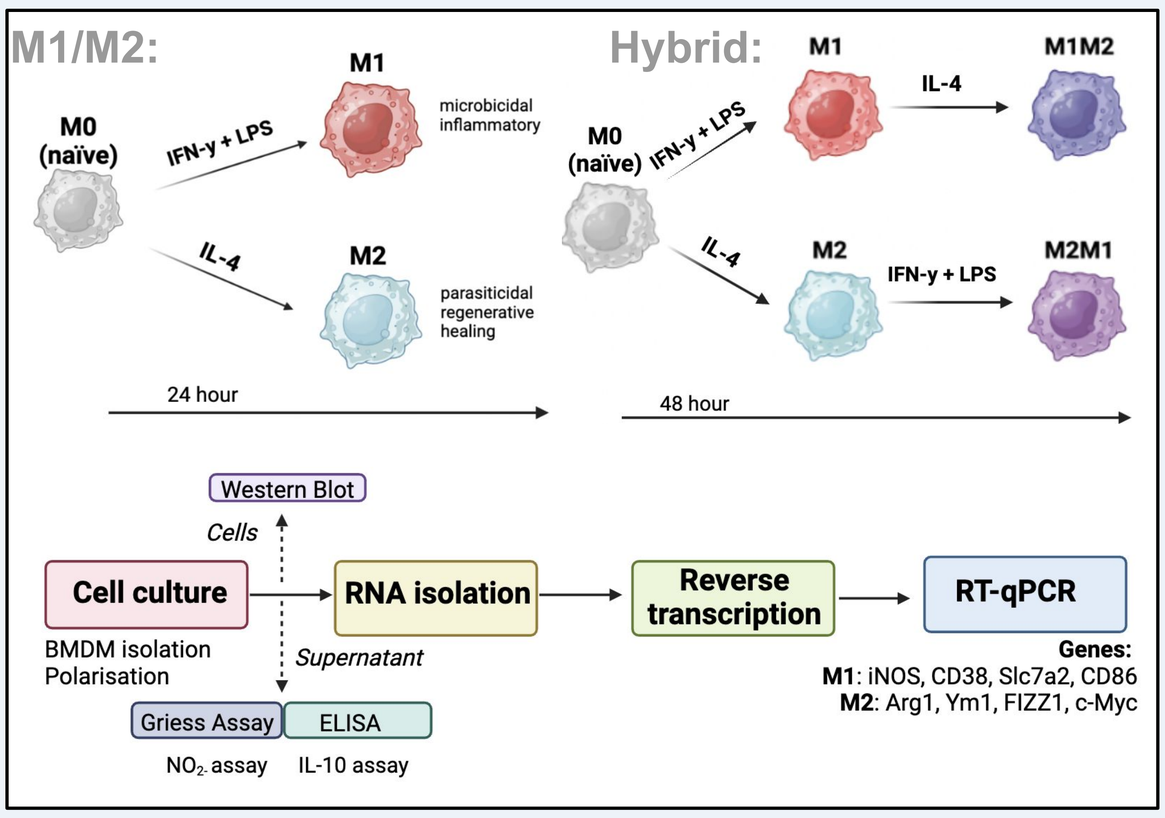
- Macrophages are key **innate immune cells**
- *In vivo,* macrophage polarisation is complex and is influenced by specific tissue microenvironment signals
- M1 and M2 macrophage models represent **extremes** of the spectrum of macrophage phenotype and function.
- **M1 phenotype:** associated with inflammatory response and microbicidal activity
- **M2 phenotype:** associated with anti-inflammatory response, parasiticidal activity and wound healing
- M1/M2 *in vitro* models provided biologically relevant foundational research for macrophage polarisation that provides starting point for *in vivo* studies

- **'Hybrid macrophage'** study provided insight into complex polarisation observed *in vivo.*
- **Spiller et al 2022, Allen J. et al 2009** studies into **human** macrophage polarisation observed that 'priming' macrophages may produce a more **robust** phenotype.
- M1 primed M2 macrophages (M1M2), observed **increased anti-inflammatory** activity and genotype was identified
- M2 primed M1 macrophages (M2M1), observed increased NO production and iNOS expression, suggesting increased **microbicidal activity**, however also **increased anti-inflammatory IL-10**
- These reports guided the research into **murine** macrophages in this study.

Aims

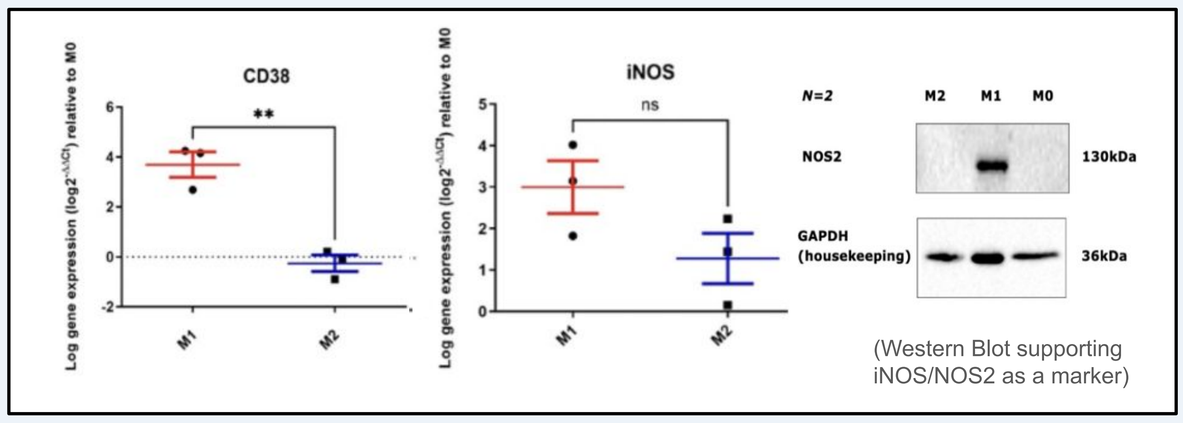
1. Identify a gene marker panel that differentiates M1 from M2 murine macrophages
2. Identify soluble markers that can provide secondary confirmation to gene analysis
3. Investigate changes in the phenotype of hybrid murine macrophages

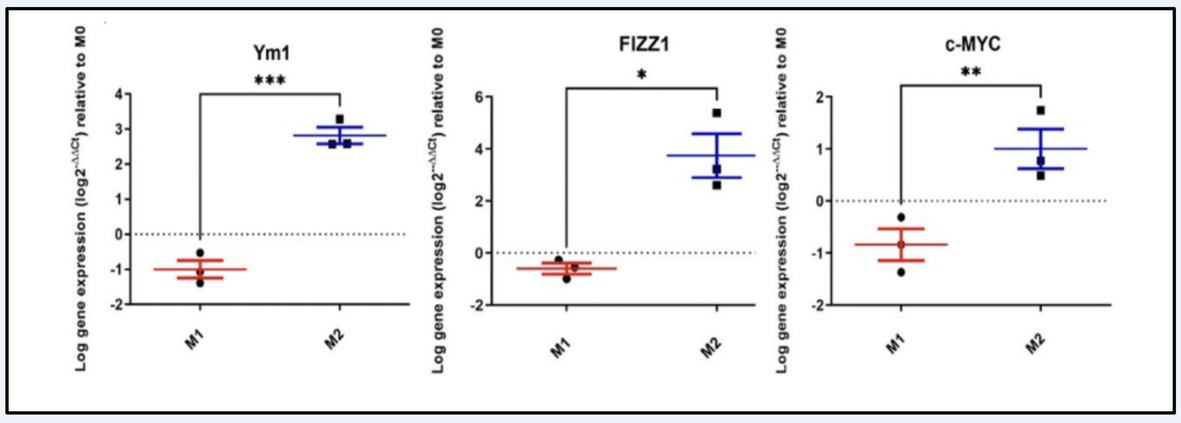
Methods



Results

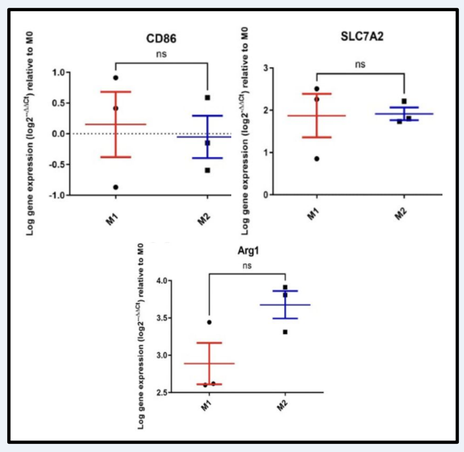
1. Gene markers

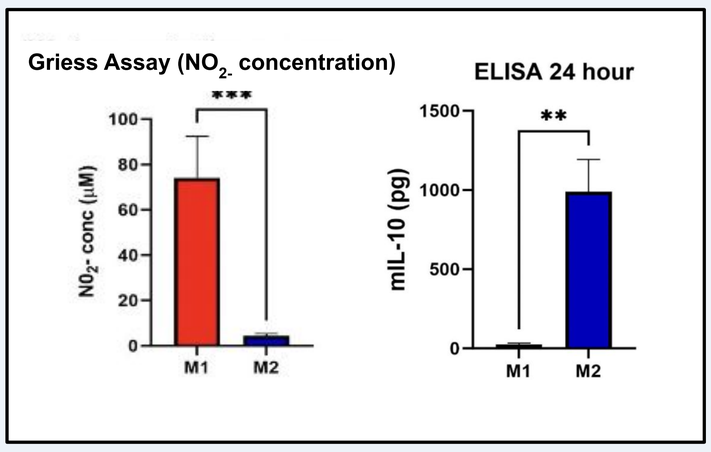




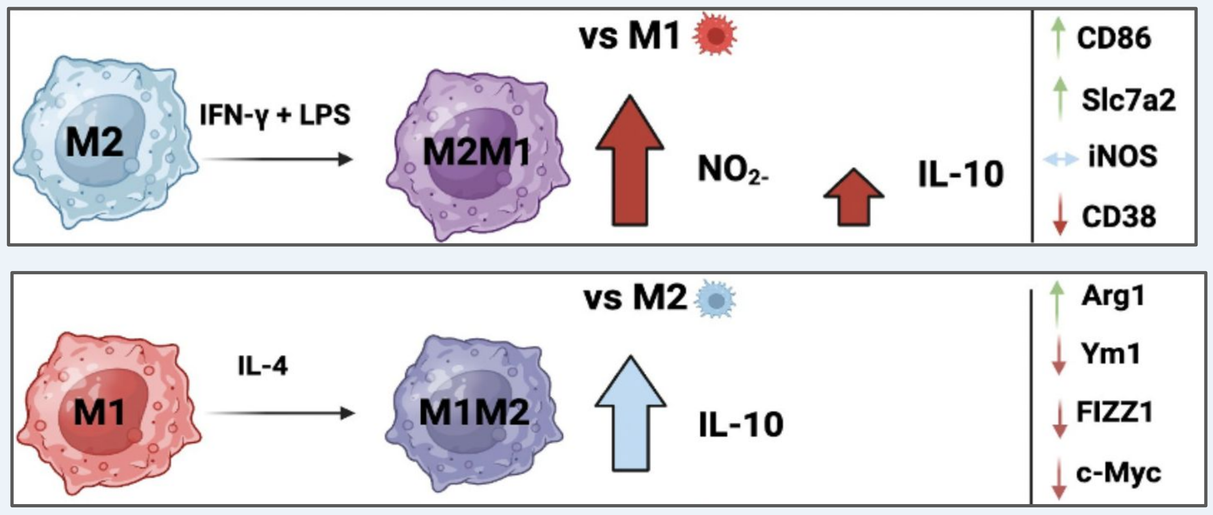
(Ineffective gene markers)

2. Soluble markers





3. Hybrid soluble marker and gene marker results summary



Future perspectives:

1. Repeat hybrid experiments with broader panel of markers
2. Use peritoneal macrophages in addition to BMDM
3. Differentiate between subsets of M2 subtype e.g. M2a, M2b etc
4. Use flow cytometry for single cell surface protein analysis

Summary

M1 markers established: **CD38, NO_{2-}** secretion (soluble)

- **iNOS** is expected to be established as definitive M1 marker with additional repeats. Western Blot and Griess Assay results indicate lack of significance observed for gene marker analysis is due to limited number of repeats carried out (n=3)
- CD86 and Slc7a2 were not identified as M1 markers

M2 markers established: **Ym1, FIZZ1, c-Myc** and **IL-10** secretion (soluble)

- Arg1 was not identified as an M2 marker

In vivo insights and preliminary findings (n=1):

- Priming macrophages can produce more robust phenotype
- Priming is already observed in wound healing in vivo (M1M2 shift)
- Hybrid M1M2 macrophages produced higher levels of IL-10 suggesting enhanced anti-inflammatory activity
- Hybrid M2M1 macrophages produced higher levels of nitrite and IL-10 suggesting enhanced microbicidal activity but some sort of protective mechanism to prevent excessive inflammation

- Gene analysis could not identify genotypes associated with M1M2/M2M1 due to lack of repeats and limited panel explored

References: Wynn TA, Chawla A, Pollard JW. Macrophage biology in development, homeostasis and disease. Nature. 2013 Apr 25;496(7446):445-55, Mylonas KJ, Nair MG, Prieto-Lafuente L, Paape D, Allen JE. Alternatively activated macrophages elicited by helminth infection can be reprogrammed to enable microbial killing. J Immunol. 2009, O'Brien EM, Spiller KL. Pro-inflammatory polarization primes Macrophages to transition into a distinct M2-like phenotype in response to IL-4. J Leukoc Biol. 2022 May;111(5):989-1000.